**7** A force $F$ acts on a body of mass $m$ for a time $t$. During this time, the velocity of the body increases from $u$ to $v$.

Which equation relates $F$, $m$, $t$, $u$ and $v$?

**A** $Fm = t(v - u)$

**B** $Fm = t(v + u)$

**C** $Ft = m(v - u)$

**D** $Ft = m(v + u)$